



Prescribed burning impacts avian diversity and disadvantages woodland-specialist birds unless long-unburnt habitat is retained[☆]



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ABSTRACT

Prescribed burning is a commonly adopted fire-management strategy that attempts to protect human life and assets by removing accumulated, flammable biomass. Heterogeneous burning patterns are often favoured in an attempt to balance fuel-reduction and biodiversity goals under the ‘pyrodiversity begets biodiversity’ paradigm. Using comprehensive spatiotemporal monitoring data, we quantified the impacts of fire on bird assemblages in the peri-urban temperate woodlands of the Mount Lofty Ranges, South Australia, where the frequency of prescribed burning is increasing. After accounting for regional trends and site effects, sites burnt 20 years previously accommodated 15% fewer birds than unburnt sites, while sites burnt in the preceding year had 22% fewer birds. Fire also modified bird assemblages, favouring generalists and ground-feeding species. Of 60 species considered, 37% were both declining and negatively impacted by recent burning, while burning reinforced increasing trends in 30% of species, particularly large, common birds (e.g., magpies, ravens, wattlebirds). Simulations of avian alpha-, beta- and gamma-diversity under different fire-management scenarios predicted higher avian diversity for scenarios that retained unburnt woodlands relative to those that managed all sites. Relative to a no-fire scenario, for example, burning sites once every 10 years was simulated to reduce the abundance of woodland generalists by 7% and woodland specialists by 10%, while retaining some long-unburnt woodland ameliorated these effects. There is a trade-off between fuel-reduction burning and conservation goals; to maximise avian diversity and avert the replacement of woodland bird species with generalists, fire-management planning should preserve long-unburnt woodland habitat.

1. Introduction

Fire is a major disturbance process that has structured global biomes, and the control of fire by humans has inevitable ecological consequences (Bowman et al., 2016). Following prevalent fire suppression in many landscapes during much of last century, the accumulation of flammable biomass and increased risk of high-intensity wildfires led to new fire-management practices (Backer et al., 2004; Schoennagel et al., 2004) including low-intensity “prescribed” burning to reduce fuel loads (Parr and Andersen, 2006; Taylor et al., 2012). Simultaneously, greater recognition of fire as a natural and ecologically

important disturbance (Allen et al., 2002) contributed to the development of patch mosaic burning strategies that sought to maximise or restore lost biodiversity by generating a mosaic of different fire histories across a landscape (Brockett et al., 2001; Parr and Andersen, 2006). The central assumption of this approach, that “pyrodiversity begets biodiversity” (Martin and Sapsis, 1992), traces back to the intermediate disturbance hypothesis (IDH) which asserts that biodiversity is maximised at moderate levels of disturbance because a range of successional stages will accommodate more species (Connell, 1978).

The contribution of ecologists to fire-management planning processes is hampered by controversy surrounding the efficacy of

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heterogeneous burning patterns (Parr and Andersen, 2006) which, in theory, should increase γ -diversity (the total species diversity across a landscape) and β -diversity (species “turnover”), while having positive or negative impacts on α -diversity (the mean diversity per site or habitat). However, empirical support for these effects is equivocal and the response of diversity metrics to fire appears to be taxon-specific (Lindenmayer et al., 2016; Pastro et al., 2011; Pastro et al., 2014; Velle et al., 2014; Whelan et al., 2002). Further, burning regimes that increase γ -, β - and/or α -diversity might be considered inappropriate when rare or threatened species are disadvantaged. In such cases, fire ecologists could aim to optimise different metrics such as the geometric mean of the species' abundances (GMA), a multiplicative index that is equally sensitive to variation in the abundance of common species as rare species (Giljohann et al., 2015; Kelly et al., 2015).

The imperative to use prescribed burning in conservation reserves is increasing as urban fringe development brings suburbia closer to protected areas (Gueneralp and Seto, 2013), and as wildfire risk increases due to more frequent extreme heat events under climate change (Westerling and Bryant, 2008; Wotton et al., 2010). However, a major challenge is to ensure that prescribed burning does not impact biodiversity negatively (Clarke, 2008; Giljohann et al., 2015; Kelly et al., 2015). Birds are conspicuous, easily monitored, and are often used as a flagship taxon in biodiversity studies (Lawton et al., 1998). Birds with specific habitat and foraging requirements respond differently to fire – some species persist in burnt areas or readily move into them, while others recolonise from source populations at different stages of vegetation succession after burning (Christensen et al., 1985; Loyn and McNabb, 2015; Russell et al., 2009). However, some bird species require or prefer habitats that have had little or no exposure to fire for extended periods (Berry et al., 2015; Kelly et al., 2015). Urban development in Australia has favoured non-native, predatory, nectarivorous and/or large native birds (Catterall et al., 2010; French et al., 2005) at the expense of small ‘specialist’ species, insectivores, and understorey and tree-cavity nesters (Ikin et al., 2014). There is an urgent need for data to guide fire management in habitat refuges that support such species (Clarke, 2008; Woinarski, 1999), particularly given that high-frequency prescribed burning can, at least initially, create open and structurally simple habitats that might favour the same birds that proliferate in urban and peri-urban environments.

These management pressures apply in the fragmented woodlands of the Mount Lofty Ranges of South Australia, a national biodiversity hotspot (Department of the Environment, 2015) that lies directly adjacent to the city of Adelaide (population c. 1.3 million). Approximately 90% of native vegetation has been cleared from the ranges since European arrival in the mid 1800's (Westphal et al., 2003) and, although some bird species are now locally extinct (Ford et al., 2001), remnant woodland ecosystems still provide an important refuge for avian biodiversity. However, there is a time-lag between habitat loss and species loss, and the abundance of many small-bodied, specialist birds continues to decline in favour of large-bodied generalist species (Szabo et al., 2011). Prescribed burning in the Mount Lofty Ranges to avert dangerous, high-intensity bushfires has also increased in frequency over the last decade (Fig. 1), a management response to loss of human life in wildfires in other parts of Australia (Bardsley et al., 2015; Gibbons et al., 2012). In 2012, the South Australian Government adopted a target to burn 5% of all public woodlands annually in high-risk areas (DEWNR, 2014), which necessitated additional burning in the Conservation Land Management Zones. The consequences of this response for avian diversity are poorly understood; however, frequent low-intensity fires could reinforce population declines in species that require a complex understorey habitat (Ford et al., 2001).

Fire management in the Mount Lofty Ranges has created a mosaic of different post-fire vegetation age classes, whilst retaining some vegetation patches that have never been burnt in recorded history (Fig. 1). Simultaneously, the Mount Lofty Ranges woodland bird monitoring programme has comprehensively monitored avian abundance and

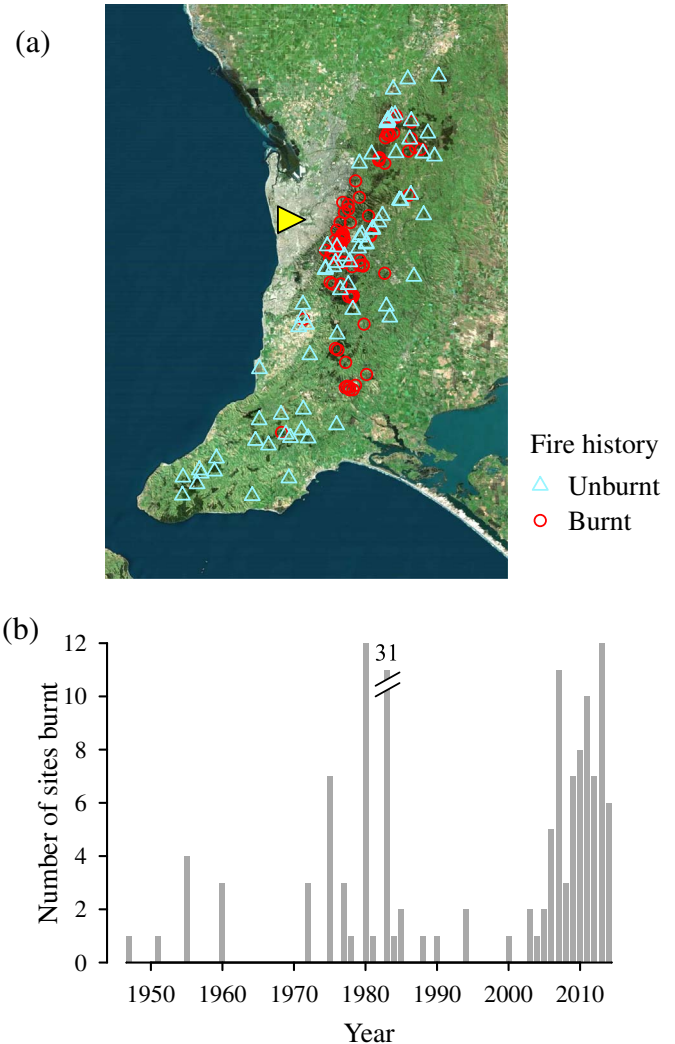


Fig. 1. (a) Bird survey sites in the Mount Lofty Ranges, South Australia, showing sites that have ($n = 81$) and have not ($n = 70$) been burnt in recorded history. The central business district of the city of Adelaide is marked with a yellow arrow. (b) The number of survey sites burnt in each year since 1947, illustrating the increased frequency of prescribed burning over the last decade. In 1983, 31 sites were burnt during the ‘Ash Wednesday’ bushfires. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

diversity at 151 woodland sites. Using this unique, spatially replicated and longitudinal dataset, we investigated the long-term responses of avian abundance and diversity to disturbance by fire that are so difficult to address with short-term experimental studies (Driscoll et al., 2010). Using multinomial diversity models fitted to the data, we also simulated α -, γ - and β -diversity under different fire-management scenarios. We predicted that avian diversity metrics should be improved at intermediate levels of fire disturbance, but maximised by management scenarios that retained some long-unburnt woodland habitat.

2. Materials and methods

2.1. Survey methodology and species selection

Avian diversity in the Mount Lofty Ranges has been surveyed annually since 2001 across a core set of 151 sites (Fig. 1a). Each site consists of a 2-ha woodland plot dominated by stringybarks (*Eucalyptus obliqua*) (83 sites) or pink gum/blue gum (*Eucalyptus fasciculosa/leucoxylon*) (68 sites). Sites are surveyed 3 times per year during Austral spring and early summer, using a standard 20-min timed search, during

which all bird species are enumerated. These bird surveys have produced data for 117 species; however, we focused on 86 native, non-migratory, woodland species for which the Mount Lofty Ranges constitute an important part of their Australian distribution (see Supplementary Table S1). We also excluded a further 26 species with a total count of < 50 individuals summed across all (> 6000) surveys, leaving 60 species for which the effects of fire could be considered robustly, while retaining 99.8% of the abundance data (Table S1).

All public woodlands in the Mount Lofty Ranges are categorised into one of three bushfire management zones: Asset Protection Zones or Bushfire Buffer Zones, within which fuel biomass is maintained below specific thresholds, and Conservation Land Management Zones which may be burnt to reduce fuel loads or for ecological purposes including the development of fire mosaics. The 151 bird survey sites are primarily found in Conservation Land Management Zones (78 sites) and on private land (67 sites), while just 6 sites fall within Bushfire Buffer Zones.

2.2. Specification of abundance and diversity models

To estimate the effects of fire on bird abundance and diversity, we controlled for trends and also site effects in our analyses. Fitting site effects accounted for between-site variation due to spatial drivers such as elevation, vegetation composition and habitat patch size. We extracted fire histories for each monitored site from spatial fire-management data layers. Of the 151 sites, 81 have been burnt in recorded history, with the first recorded fire at a site occurring in 1947. The bird surveys cover the recent period over which there has been an increase in prescribed burning (Fig. 1b) and, since 2001, 53 sites have been burnt at least once. We treated the three survey sessions as replicate observations for each site and year combination.

To consider how avian abundance and diversity respond to fire, we calculated the time since the last fire event (*TSLF*; years) for each site in each year of sampling. This covariate could not be calculated for surveys from sites that had never been burnt, so we constructed model matrices to fit the effect of a binary ‘fire history’ factor (*fire*: never burnt/burnt previously) and, conditional on a site having been burnt, the effect of the continuous *TSLF* covariate. Because the impacts of fire are likely to diminish over time, we also tested different model specifications that limited fire effects to occur within a specific timespan *X*, where *X* was varied between 5 and 50 years. For these models, we re-defined fire status (*fireX*: not burnt or burnt within the last *X* years) and time since the last fire (*tslfX*: time to the last fire within the last *X* years).

2.3. Abundance models

We used generalised additive mixed models (GAMMs) to investigate non-linear temporal trends and the effect of fire on avian abundance (i.e., the total number of birds observed per site). We fitted these abundance models in the R computing environment (R Development Core Team, 2014) using functions within the package *mgcv* (Wood, 2004), and assumed a negative-binomial distribution for these count data which we selected over alternative distributions based on Akaike's information criterion (AIC). Specifically, we fit two models: (1) a model including a trend term and random site effects only; and (2) a model that additionally included time-since-last-fire as a covariate, within which the timespan over which the effects of the last fire was felt was selected by AIC (see Supplementary Table S2).

2.4. Multinomial diversity models

We used multinomial diversity models (MDMs) to quantify simultaneously the effect of multiple covariates on avian diversity and the proportional abundance of each bird species (De'ath, 2012). The MDM is a multinomial regression that links covariates to the proportional abundance of each species within an assemblage. One advantage of the MDM is that it can identify covariates that contribute to species

turnover (i.e., affect proportional abundances) but do not have strong impacts on α -diversity (De'ath, 2012). Further, the proportional-abundance responses of individual species can be aggregated to higher levels (e.g., foraging guilds) and also used to calculate α -, γ - and β -diversities or other univariate summary statistics such as the geometric mean of the species abundances (Kelly et al., 2015). All MDMs were fitted in R using the package *MDM* (De'ath, 2013).

As above, we fit MDMs that: (1) included trends and site effects only; and (2) additionally included fire effects after an optimum fire-effect specification had been selected. Given we modelled diversity in an assemblage comprising 60 species, to guard against overfitting we selected the optimum fire-effects timespan using leave-one-year-out cross-validation. Specifically, we subset the 14 years of data to produce training (13 years) and validation datasets (1 year), and evaluated the performance of MDMs fitted to the training data by calculating the mean predictive deviance across the hold-out sets. Similarly, we compared the ability of MDMs to predict the presence or absence of bird species accurately, by generating 1000 multinomial realisations (i.e. predicted species counts) for the hold-out surveys of known abundance, and calculated the mean cross-validation misclassification rate. We used the selected MDM to consider species-level responses to fire, but also aggregated responses to the level of three specialisation groups (generalists, woodland generalists and woodland specialists) using an *a priori* classification (Attwood et al., 2009) as well as the level of seven foraging guilds (Supplementary Table S1). We used a spatial block bootstrap to generate 95% confidence intervals on the fitted trends and time-since-last-fire effects. To do so, we performed 1000 bootstrap replicates by resampling sites with replacement using a circular spatial block (radius = 10 km) and refitting the MDM to bird data from each bootstrap sample. Resampling using spatial blocks accounts for spatial dependence in the data (Plant, 2012) and should produce confidence intervals that are less affected by spatial variation in prescribed-burning practices.

2.5. Fire-management scenario testing

To explore how the frequency of woodland disturbance by prescribed burning is expected to affect bird diversity in the Mount Lofty Ranges, we coupled the selected abundance model and MDM to simulate bird assemblages at the surveyed sites assuming different prescribed-burning strategies over the historical period between 2001 and 2014. Specifically, we tested nine hypothetical scenarios: no fires over the entire period; burning of 5, 10, 15 and 20% of sites per year; and burning of 5, 10, 15 and 20% of sites per year but only managing those sites that had a fire history in 2001. The latter four scenarios left 94 sites with no fire history undisturbed by fire for the whole period. For each scenario that implemented prescribed burning, we assumed rotational burning of sites through time (e.g., assuming 10% of sites burnt annually, a site burnt in one year would not be burnt for the following 9 years), and performed 500 stochastic replicates to account for random sampling of the sites selected for management using fire. Using average results for the final year simulated, we compared these scenarios by calculating α -, γ - and β -diversity, the geometric mean of the species' abundances (GMA), and the mean abundance per site for different specialisation groups.

3. Results

Woodland burning had complex effects on bird abundance and diversity. Before considering fire effects, a GAMM model of bird counts over time demonstrated that bird abundance in the Mount Lofty Ranges declined non-linearly (estimate [95% confidence interval]) from a peak of 35.7 [33.3, 38.0] birds site⁻¹ in 2001 to 19.5 [18.2, 20.8] birds site⁻¹ in 2014, which is equivalent to a 45% decline in abundance over this period (Fig. 2a). Including a fire-effect term in this model that incorporated the full range of time-since-last-fire values was supported

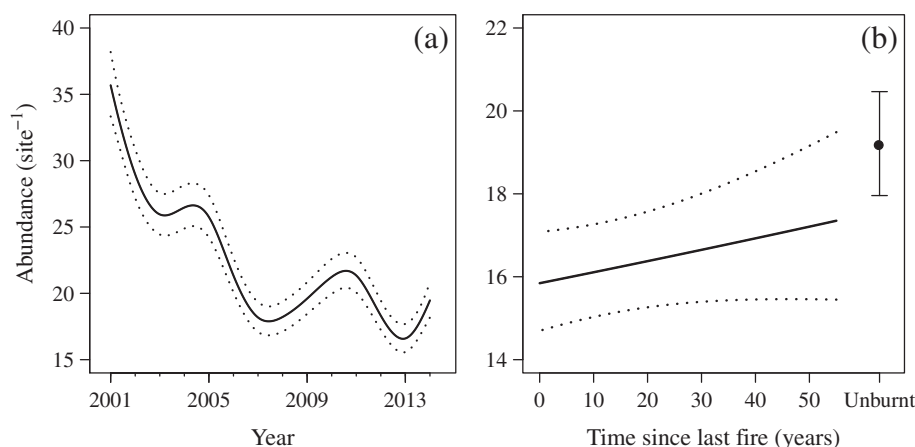


Fig. 2. Bird abundance patterns derived from generalised additive mixed models that accounted for site effects: (a) the trend in abundance over time; and (b) the effect of fire on abundance, after correcting for the non-linear trend. Solid lines represent the partial effects and dotted lines denote 95% confidence intervals.

based on AIC (10,807 cf. 10,832, $\Delta\text{AIC} = 24.3$; see Supplementary Table S2 for AIC-based comparisons of different fire-effect specifications). Using this model formulation and conditioning on the mean year of sampling, we estimated that bird abundance was 19.2 [18.0, 20.5] birds at sites that have never been burnt in recorded history, while abundance declined from 17.4 [15.4, 19.5] birds at sites burnt 55 years ago to 15.8 [14.7, 17.1] birds at sites burnt in the year immediately before surveys (Fig. 2b). Bird abundance at sites that had not been burnt in recorded history was clearly different from that at sites burnt 20 years ago (16.4 [15.3, 17.6] birds site⁻¹) or more recently (Fig. 2b).

Species richness and α -diversity calculated directly from the raw data did not reveal strong trends in diversity over time (Fig. 3a), and this conclusion was supported by parametric estimates of α -diversity derived from a multinomial diversity model (MDM) that fitted site and trend terms only (Fig. 3c). We then incorporated time-since-last-fire predictors that limited the influence of fire on species proportional abundances to a maximum timespan of 20 years, because this specification produced the lowest MDM deviance as well as the lowest cross-validation deviance and misclassification rate (see Supplementary Table S3). In this case, truncating the time-since-last-fire covariate improved the model metrics by reducing the leverage of the relatively few surveys done at sites that were last burnt > 20 years in the past.

Again, this selected MDM demonstrated no relationship between parametric α -diversity and the time since the last fire event (Fig. 3d).

The stability of avian α -diversity over time and the post-fire interval hides important species- and group-level trends and fire effects revealed by the MDMs. The proportional abundance of generalist species approximately doubled between 2001 and 2014 (0.034 to 0.069) and, after accounting for this trend, the representation of generalists was more than twice as great at sites burnt 20 years ago than at sites burnt in the preceding year (0.075 cf. 0.035) (Fig. 4a(i)). Over the same period, the proportional abundance of woodland generalists increased (0.409 to 0.506) while woodland specialists decreased (0.557 to 0.426) (Fig. 4a(ii–iii)). The representation of woodland specialists was 12% higher at sites burnt 20 years ago than at sites burnt in the preceding year (0.522 cf. 0.468); however, confidence intervals on these partial responses (derived from the spatial bootstrap) were wide. After aggregating to the foraging-guild level, the clearest trend in proportional abundance was a doubling for carnivorous birds that feed primarily on the ground (0.045 to 0.093) (Fig. 4b(i)). Weaker increases (for ground granivores) and decreases (for ground insectivores and nectarivore-insectivores) were also observed (see Supplementary Fig. S1 for response curves for all seven foraging guilds). After accounting for these trends, we found only one clear guild-level effect of fire – a strong positive

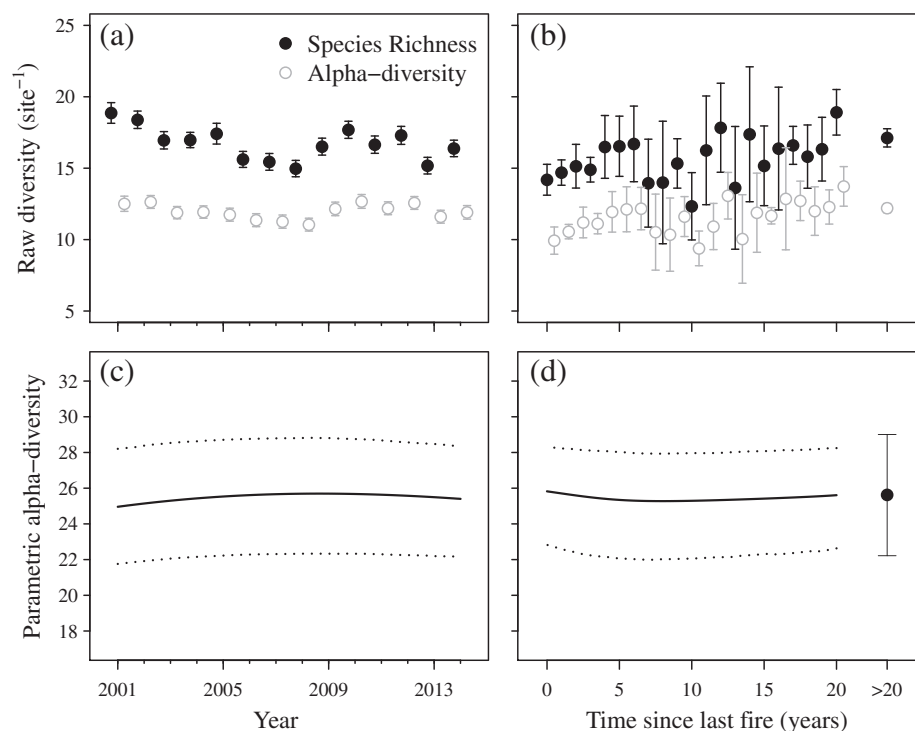


Fig. 3. Avian species richness and α -diversity patterns. (a, b) Mean species richness and α -diversity derived from the raw data, as a function of time (a) and time since the last fire (b). Error bars represent standard errors calculated at the site-level. (c, d) Parametric α -diversity derived from multinomial diversity models for: (c) a model that fit site effects and a trend term; and (d) a model that additionally included fire effects (i.e., panels illustrate estimated effects of fire on diversity after correcting for the overall trend). Solid and dotted lines denote the mean response and 95% confidence intervals, respectively, calculated from 1000 spatial-block bootstrap replicates. In panels (b) and (d), the classification of sites that have not been burnt within the last 20 years includes sites that have never been burnt in recorded history. Note that parametric α -diversity exceeds that calculated directly from the data and can be viewed as the diversity expected at a site with very high bird abundance.

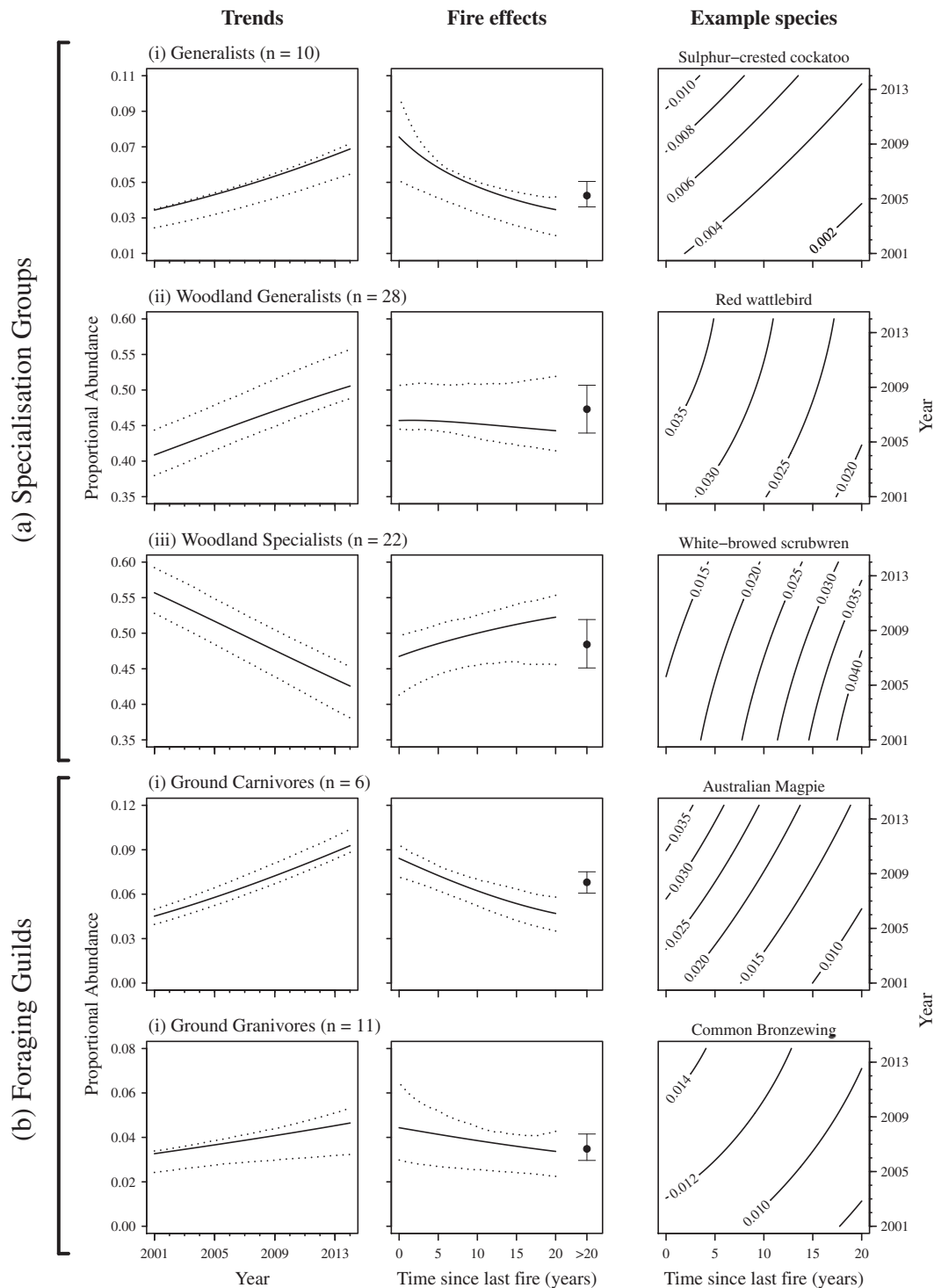


Fig. 4. Proportional-abundance responses at the level of (a) specialisation groups and (b) foraging guilds. The first column presents trends over time from a multinomial diversity model that fitted site effects and a trend term. The second column is derived from a model that additionally included fire effects and shows the change in proportional abundance as a function of time since the last fire (i.e., estimated fire effects after correcting for the overall trends). Contour plots in the final column show estimated proportional abundances for one example species from each group, as a function of year and time since the last fire.

effect of recent burning on the proportional abundance of ground carnivores (Fig. 4b(i)).

To consider how an increased frequency of prescribed burning is expected to influence avian conservation goals, we classified species into four groups according to species-specific trends in proportional abundance (decreasing/increasing) and the effects of burning within the preceding year relative to no burning within the last 20 years (negative/positive after correcting for overall trends). The largest group

comprised 37% of species for which recent burning is expected to reinforce decreasing trends, and included many small birds such as the silvereye (*Zosterops lateralis*) and superb fairy-wren (*Melurus cyaneus*) (Table 1). The second largest group comprised 30% of species for which anthropogenic burning is expected to consolidate increasing trends, including many large birds such as the Australian magpie, little raven (*Corvus mellori*) and grey currawong (*Strepera versicolor*) (Table 1).

Simulating hypothetical fire-management regimes over the period

Table 1

Classification of species according to trends in proportional abundance (decreasing/increasing) and the effects of burning within the preceding year relative to no burning within the last 20 years (negative/positive after correcting for overall trends). Shading illustrates the certainty with which each species is placed in one of the four cells: ■ > 95% bootstrap support for both the trend and fire effect; ■ > 95% support for one effect and 75–95% for the other; □ 75–95% support for both effects; □ < 75% support for both effects. Numbers in brackets indicate the relative change in proportional abundance of each species between 2001 and 2014. Abbreviations for specialisation groups are: G, generalist; WG, woodland generalist; WS, woodland specialist. Abbreviations for foraging guild classifications are: AI, arboreal insectivore; F, frugivore; GC, ground carnivore; GG, ground granivore; GI, ground insectivore; N-I, nectarivore-insectivore; P, psittaciforme.

		EFFECT OF FIRE			
		Negative	Positive		
TREND	Decreasing	White-naped honeyeater (-0.71) [WS; N/I]	Noisy miner (-0.92) [G; N/I]	Decreasing	
		Magpie lark (-0.71) [G; GC]	Spotted pardalote (-0.73) [WS; AI]		
		White-plumed honeyeater (-0.69) [WG; N/I]	Mistletoebird (-0.68) [WS; F]		
		Silvereye (-0.66) [WG; F]	Brown-headed honeyeater (-0.61) [WS; N/I]		
		Brown thornbill (-0.55) [WS; AI]	Willie wagtail (-0.60) [G; GI]		
		Purple-crowned lorikeet (-0.51) [WG; P]	Striated thornbill (-0.56) [WS; AI]		
		White-browed scrubwren (-0.45) [WS; GI]	Crested pigeon (-0.52) [G; GG]		
		White-winged triller (-0.43) [WG; AI]	Red-browed finch (-0.33) [WG; GG]		
		Eastern spinebill (-0.36) [WG; N/I]	Buff-rumped thornbill (-0.21) [WS; GI]		
		Red-rumped parrot (-0.29) [WG; GG]	Yellow-faced honeyeater (-0.17) [WS; N/I]		
		Superb fairy-wren (-0.29) [WS; GI]	Yellow-rumped thornbill (-0.02) [WG; GI]		
		Musk lorikeet (-0.22) [WG; P]	Weebill (-0.01) [WS; AI]		
		Varied sittella (-0.21) [WS; AI]	Black-faced cuckoo-shrike (0.06) [WG/AI]	Increasing	
		Elegant parrot (-0.21) [WG; GG]	White-browed babbler (0.09) [WS; GI]		
	Increasing	Crescent honeyeater (-0.19) [WS; N/I]	Adelaide rosella (0.15) [WG; P]		
		Yellow thornbill (-0.18) [WS; AI]	Yellow-tailed black-cockatoo (0.25) [WG; P]		
		New holland honeyeater (-0.14) [WG; N/I]	Red wattlebird (0.32) [WG; N/I]		
		Brush bronzewing (-0.13) [WS; GG]	Common bronzewing (0.39) [WG; GG]		
		Horsfield's bronze-cuckoo (-0.09) [WG; AI]	Galah (0.42) [G; GG]		
		Dusky woodswallow (-0.07) [WG; AI]	Grey shrike-thrush (0.67) [WG; AI]		
		Peaceful dove (-0.06) [WG; GG]	Striated pardalote (0.71) [WG; AI]		
		Golden whistler (-0.04) [WS; AI]	Laughing kookaburra (0.71) [WG; GC]		
		Sacred kingfisher (0.18) [WG; GC]	Scarlet robin (0.74) [WS; GI]		
		White-winged chough (0.19) [WG; GI]	Australian magpie (1.09) [G; GC]		
		Grey fantail (0.32) [WS; AI]	Grey currawong (1.11) [WG; GC]		
		White-throated treecreeper (0.68) [WS; AI]	Little raven (1.36) [G; GC]		
		Rainbow lorikeet (0.82) [WG; P]	Rufous whistler (1.48) [WS; AI]		
		Fan-tailed cuckoo (1.28) [WS; AI]	Tree martin (1.51) [WG; AI]		
		Shining bronze-cuckoo (2.12) [WG; AI]	Eastern rosella (1.73) [G; GG]		
		Little corella (6.86) [G; GG]	Sulphur-crested cockatoo (2.40) [G; GG]		

2001 to 2014 revealed differences between scenarios in the avian diversity expected in the final year but provided no support for the IDH (Fig. 5a). As the percentage of woodland sites burnt each year was increased, scenarios that retained unburnt woodlands produced higher α - and γ -diversities and also GMA than those that managed all sites (Fig. 5a). Increased fire frequency also resulted in fewer birds per site, increasing the abundance of generalist birds while lowering the abundance of woodland generalists and specialists, but again leaving unburnt sites unmanaged had a protective effect (Fig. 5b). Relative to a no-fire scenario, for example, the simulated burning of 10% of sites

each year (i.e. equivalent to burning each site once every 10 years) reduced the abundance of woodland generalists and specialists by 7 and 10%, respectively, equivalent to a decline of 1.7 woodland birds site⁻¹. However, retaining unburnt sites while burning the same number of sites per year caused a reduction of 0.3 woodland birds site⁻¹ only (Fig. 5b[iii]).

4. Discussion

In remnant vegetation adjacent to urban populations, the contention

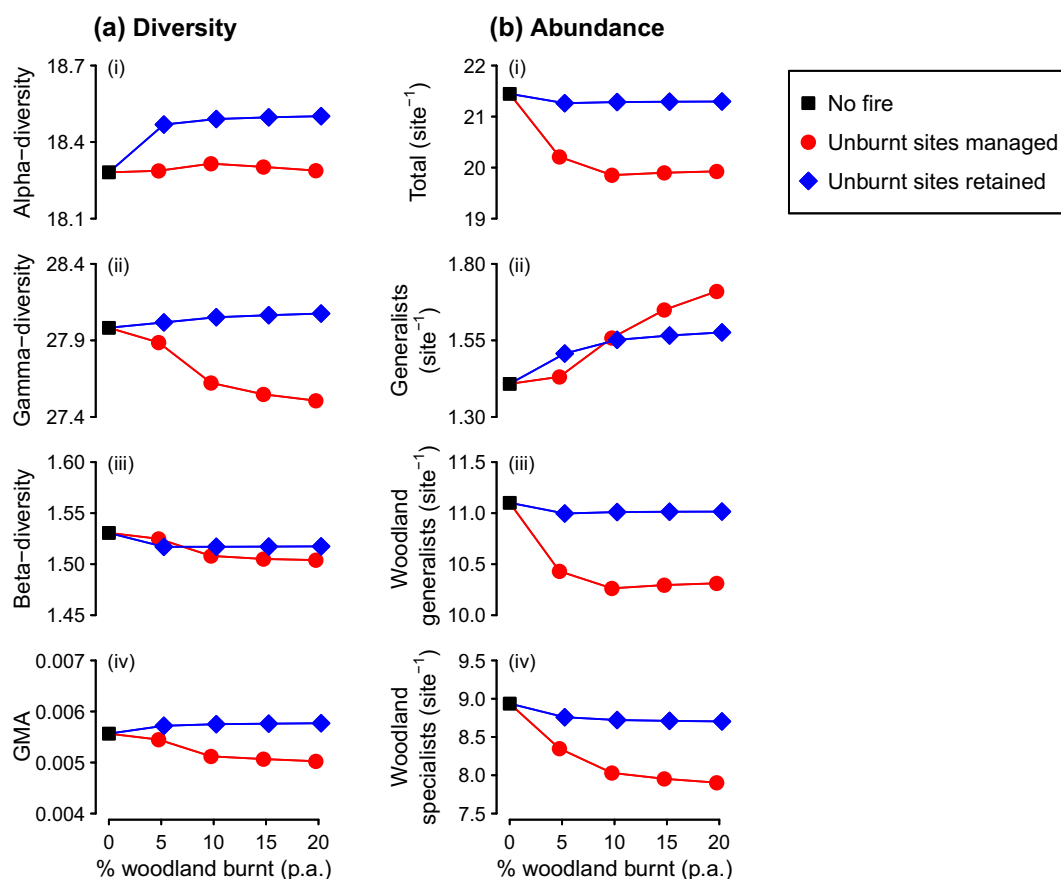


Fig. 5. Simulated avian diversity and abundance for the year 2014, assuming different hypothetical fire-management scenarios operating over the period between 2001 and 2014. (a) Avian diversity metrics calculated were α -, γ - and β -diversities (i–iii) and the geometric mean abundance (GMA; iv). (b) The mean abundance per site for: (i) all species, (ii) generalists, (iii) woodland generalists, and (iv) woodland specialists.

that prescribed-burning regimes can restrict fuel loads while safeguarding biodiversity and conserving rare or threatened species has been hotly debated (Gill and Williams, 1996). The intermediate disturbance hypothesis, which posits that diversity is maximised at moderate levels of disturbance that inhibit species' dominance (Connell, 1978), suggests that if moderate fire disturbance can suppress fuel loads then these two goals could be achieved simultaneously. Our simulations indicated that, unless long-unburnt habitats are protected, avian diversity metrics are likely to be suboptimal (Fig. 5). Disturbing all habitats with a fixed fire frequency, such as burning 5% of woodlands per year as currently mandated for the Mount Lofty Ranges, is therefore unlikely to produce beneficial diversity outcomes, at least for birds. A key component of the 'pyrodiversity begets biodiversity' paradigm is the creation of different post-fire vegetation age classes (Kelly et al., 2015; Martin and Sapsis, 1992), and the retention of long-unburnt woodland habitat is an essential component of this approach, and likely critical for maintaining bird diversity in the face of ongoing urbanisation pressures.

One proposed mechanism for the IDH is that moderate disturbance regimes can accommodate rapidly colonising, competitively inferior species that quickly disperse into disturbed habitats, as well as competitively superior species that eventually replace them (Connell, 1978; Fox, 2013). The immediate effect of prescribed burning is to remove the understorey and expose the ground surface, thereby producing structurally simple habitats which potentially favour colonisers and/or generalist bird species. Although our results did not support the IDH, they demonstrated a strong positive relationship between recent burning and the proportional abundance of generalists (Fig. 4a(i)). Of the ten species we classified as generalist birds (Attwood et al., 2009), nine primarily forage on the ground and could therefore be advantaged

by recent burning. At the guild-level, the clearest effect of fire was to favour large, ground carnivores, including the Australian magpie, little raven, and grey currawong (Fig. 4, Table 1). This is concerning given that the proportional abundance of ground carnivores has more than doubled in the Mount Lofty Ranges since 2001 (Fig. 4b(i)), and many of these species are already common in peri-urban habitats. Furthermore, these species display aggressive, predatory behaviour and could potentially affect the survival and breeding success of other bird species.

Small-bodied and insectivorous birds have declined in Australia's fragmented suburban environments (Catterall et al., 2010; French et al., 2005) but persist in vegetated areas with complex understorey structure and composition (Ikin et al., 2014). Our results suggested that frequent prescribed burning in the Mount Lofty Ranges could be a threatening process for 22 species that are already declining across the region (Table 1; Szabo et al., 2011), including small-bodied species such as the white-naped and crescent honeyeaters (*Melithreptus lunatus* and *Phylidonyris pyrrhopterus*), the white-browed scrubwren (*Sericornis frontalis*) and the silvereye (*Zosterops lateralis*). Although these species have very different foraging requirements, it is possible that the indirect effects of fire are negatively affecting these birds by favouring colonisation by large and/or aggressive birds that prosper in open environments.

In this study, we focused solely on the impacts of variation in post-fire age classes. Unfortunately, due to data limitations we were unable to consider fire intensity which is another key component of pyrodiversity that determines the degree of canopy scorched (Tingley et al., 2016) and impacts plant regeneration and recruitment post-fire (Vivian et al., 2008). We also assumed that the site-level effects of fire could be aggregated and used to produce regional estimates of avian diversity and abundance under different hypothetical burning regimes. This approach neglects the possible importance of the proportion of

habitats burnt at local (e.g., areas around 2-ha bird survey sites) and larger scales. However, preliminary testing of additional fire covariates (e.g., the proportion of woodland burnt or mean post-fire age within buffer regions placed around survey sites, and the average number of burns at each site within different time intervals) produced no improvement in model selection criteria over the time-since-last-fire covariate (T. Prowse, unpublished data). We also suggest that, given the highly fragmented nature of remnant woodlands of the Mount Lofty Ranges, the primary impacts of prescribed burning will be relatively confined in space. We note too that our data-driven simulations might underestimate the impact of prescribed burning on rare woodland birds which might have limited dispersal opportunities following the application of fire (Whelan et al., 2002).

Although recent research supports the notion that pyrodiversity improves faunal diversity in the conifer forests of North America (Ponisio et al., 2016; Tingley et al., 2016), our study adds to the growing body of Australian literature indicating that heterogeneous burning patterns do not necessarily beget higher diversity. For example, pyrodiversity has negligible to limited effects on the diversity of reptiles and birds in the Australian mallee (Farnsworth et al., 2014; Kelly et al., 2015; Taylor et al., 2012). However, the number of rare bird species is strongly correlated with the amount of older vegetation in mallee landscapes (Taylor et al., 2012). These results, and the results of our study, highlight the drawback that univariate measures of species diversity do not consider species composition explicitly.

In the Mount Lofty Ranges, for example, increased fire frequency is expected to modify the bird assemblage but something of a balance between increasingly abundant generalists and decreasing woodland species ensures that α -, γ - and β -diversities, as well as the geometric mean of the species' abundances, are unlikely to change dramatically due to fire management (Fig. 5). We argue that, in remnant woodlands such as these, the development of fire-management strategies should seek to avoid reinforcing the biotic homogenisation that is occurring in urban and peri-urban environments (Ikin et al., 2014; McKinney, 2006; Olden et al., 2004; Taylor et al., 2013). Fire regimes that retain unburnt woodlands of sufficient size and distribution to permit the development of complex understory habitats are likely to be most beneficial for small woodland birds (Barth et al., 2015; Ford et al., 2001) and less suitable for large, aggressive species that are already common.

It is therefore crucial that all prescribed burning implemented is clearly necessary to reduce bushfire risk. Frequently burning strategic areas, such as the Asset Protection and Bushfire Buffer Zones of the Mount Lofty Ranges, can consistently restrict flammable biomass adjacent to vulnerable or populated regions (Gibbons et al., 2012). However, it is debatable whether broad-scale burning targets, such as the commitment to burn 5% of all high-risk woodlands in the Mount Lofty Ranges per year, are useful. For example, such low-frequency burning might have little impact on the extent of moderate-to-high intensity unplanned fires (Cary et al., 2017; Price et al., 2015). We suggest that an annual woodland-burning target of 5% represents sufficient disturbance to compromise avian diversity if applied indiscriminately (Fig. 5) and is probably too infrequent to limit the extent of subsequent bushfires because fuel loads recover quickly.

In Australia, 50% of nationally listed threatened species occur in vulnerable urban fringe habitats (Bekessy et al., 2012). Poor land-use planning has allowed urban expansion into regions of high biodiversity value that are also prone to fire risk (Rayner et al., 2015), leading in many cases to fire-threat mitigation and associated biodiversity losses (Bardsley et al., 2015; Gibbons et al., 2012). Our results support the notion that development planning regulations should be adjusted to limit new developments in high fire-risk zones to restrict future prescribed-burning requirements. Without changes to current human development trends in the Mount Lofty Ranges, for example, the increasing social and political imperative for fuel-load reduction will risk the continuing decline and local extinction of more woodland bird species.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biocon.2017.09.005>.

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